



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

**SENIOR CERTIFICATE EXAMINATIONS/
SENIORSERTIFIKAAT-EKSAMEN
NATIONAL SENIOR CERTIFICATE EXAMINATIONS/
NASIONALE SENIORSERTIFIKAAT-EKSAMEN**

**MATHEMATICS P2/
WISKUNDE V2**

MARKING GUIDELINES/NASIENRIGLYNE

2021

**MARKS: 150
PUNTE: 150**

**These marking guidelines consist of 23 pages.
*Hierdie nasienriglyne bestaan uit 23 bladsye.***

NOTE:

- If a candidate answers a question TWICE, only mark the FIRST attempt.
- If a candidate has crossed out an attempt of a question and not redone the question, mark the crossed out version.
- Consistent accuracy applies in ALL aspects of the marking memorandum. Stop marking at the second calculation error.
- Assuming answers/values in order to solve a problem is NOT acceptable.

LET WEL:

- As 'n kandidaat 'n vraag TWEE KEER beantwoord, sien slegs die EERSTE poging na.
- As 'n kandidaat 'n antwoord van 'n vraag doodtrek en nie oordoen nie, sien die doodgetrekte poging na.
- Volgehoue akkuraatheid word in ALLE aspekte van die memorandum toegepas. Hou op nasien by die tweede berekeningsfout.
- Aanvaar van antwoorde/waardes om 'n probleem op te los, word NIE toegelaat nie.

GEOMETRY	
S	A mark for a correct statement (A statement mark is independent of a reason)
	<i>'n Punt vir 'n korrekte bewering ('n Punt vir 'n bewering is onafhanklik van die rede)</i>
R	A mark for the correct reason (A reason mark may only be awarded if the statement is correct)
	<i>'n Punt vir 'n korrekte rede ('n Punt word slegs vir die rede toegeken as die bewering korrek is)</i>
S/R	Award a mark if statement AND reason are both correct
	<i>Ken 'n punt toe as die bewering EN rede beide korrek is</i>

QUESTION/VRAAG 1

1.1

26	13	3	18	12	34	24	58	16	10	15	69	20	17	40
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1.1.1(a)	$\bar{x} = \frac{375}{15}$ $\bar{x} = 25 \text{ MB}$ Answer only: Full marks	✓ 375 ✓ answer (2)
1.1.1(b)	$\sigma = 17,65 \text{ MB}$	✓ answer (1)
1.1.2	$25 + 17,65 = 42,65$ $\therefore 2 \text{ days}$	✓ 42,65 ✓ 2 (2)
1.1.3	Overall $\bar{x} = \frac{80}{100} \times 25$ $= 20 \text{ MB}$ $\frac{375 + x}{30} = 20$ $x = 600 - 375$ $= 225$ maximum total amount of data that Sam must use for the remainder of the month: 225 MB	✓ Overall $\bar{x} = 20$ ✓ $\frac{375 + x}{30} = 20$ ✓ answer (3)

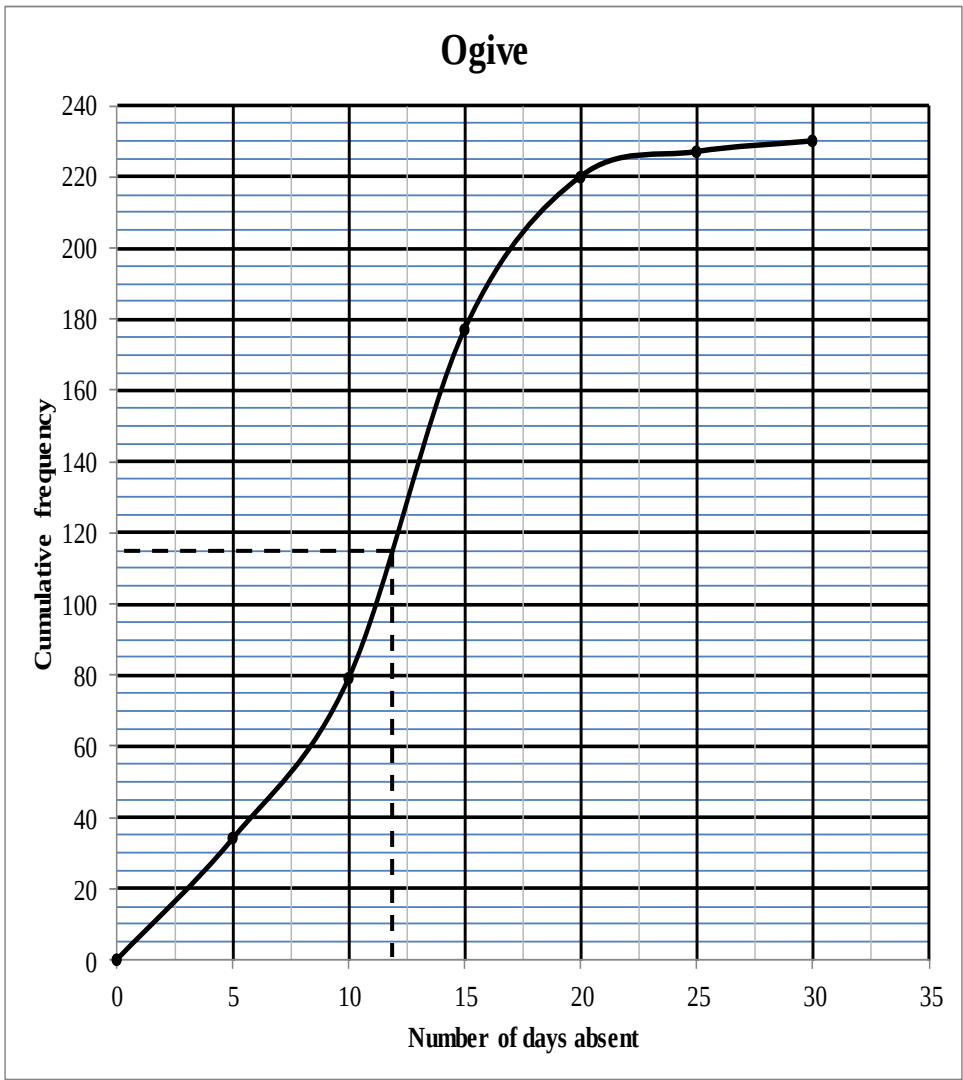
1.2

Wind speed in km/h (x)	2	6	15	20	25	17	11	24	13	22
Temperature in °C (y)	28	26	22	22	16	20	24	19	26	19

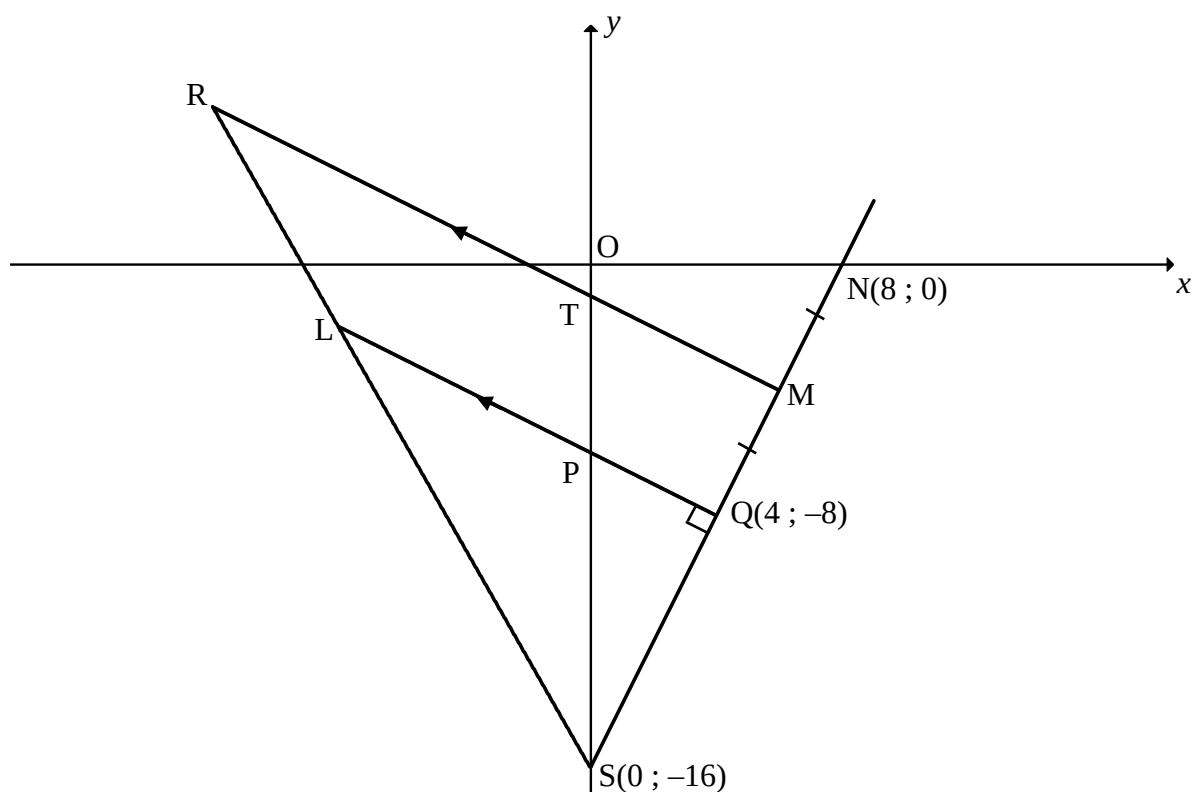
1.2.1	$a = 29,35$ $b = -0,46$ $\hat{y} = 29,35 - 0,46x$	✓ a ✓ b ✓ equation (3)
1.2.2	$y = 25,20 \text{ °C}$ (calculator) OR $\hat{y} = 29,35 - 0,46(9)$ $y = 25,21 \text{ °C}$	✓✓ answer (2) ✓ substitution ✓ answer (2)
1.2.3	$b < 0$, indicating that as the wind speed increases the temperature decreases.	✓ interpretation (1)
[14]		

QUESTION/VRAAG 2

Number of days absent	Number of learners	Cumulative frequency
$0 \leq x < 5$	34	34
$5 \leq x < 10$	45	79
$10 \leq x < 15$	98	177
$15 \leq x < 20$	43	220
$20 \leq x < 25$	7	227
$25 \leq x < 30$	3	230

2.1	Modal class: $10 \leq x < 15$	✓ answer (1)
2.2	177 learners	✓ answer (1)
2.3	230 learners	✓ answer (1)
2.4	 Ogive Cumulative frequency Number of days absent	✓ grounding at (0; 0) ✓ shape ✓ upper limits ✓ All other points correct (4)
2.5	The median is at position 115. ☐ value of median is 12 days (accept 11 – 14) Answer only: Full marks	✓ reading off at 115 ✓ answer (2)
[9]		

QUESTION/VRAAG 3

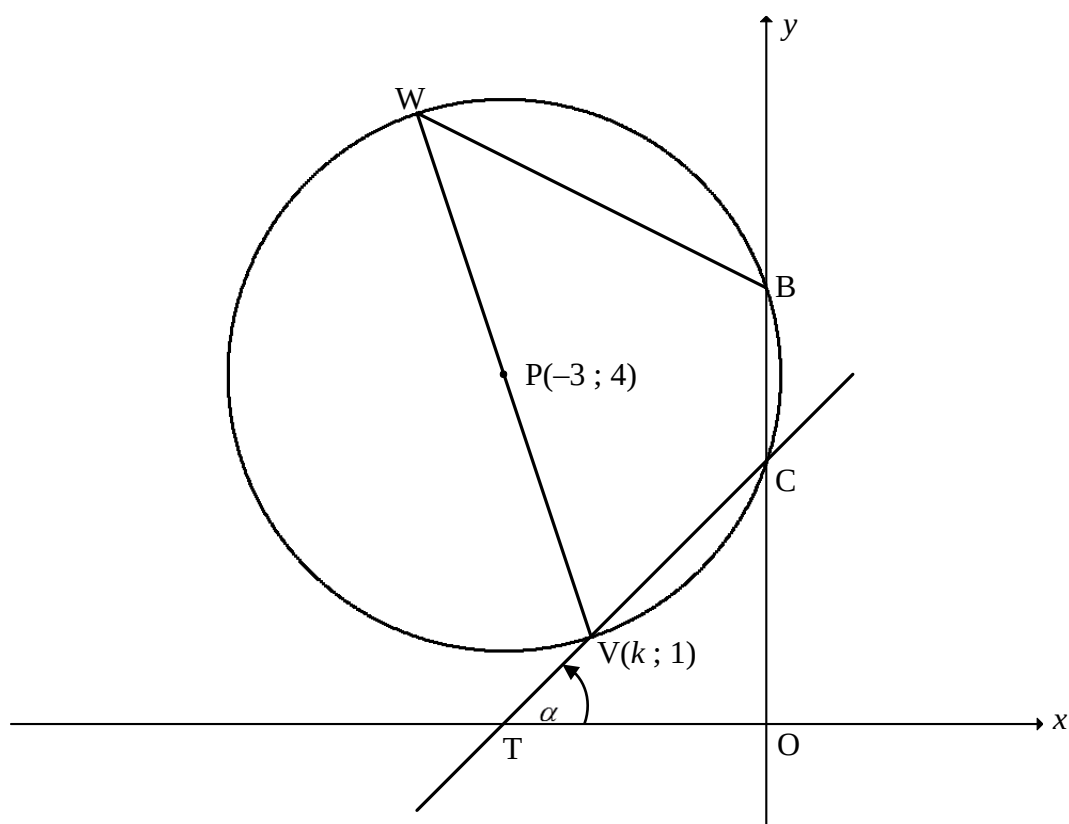


3.1	$M\left(\frac{4+8}{2}; \frac{-8+0}{2}\right)$ $M(6; -4)$	✓ x_M ✓ y_M (2)
3.2	$m_{NS} = \frac{0 - (-16)}{8 - 0} \text{ or } m_{NQ} = \frac{0 - (-8)}{8 - 4} \text{ or } m_{QS} = \frac{-8 - (-16)}{4 - 0}$ $= 2$	✓ subst N and Q or N and Q or Q and S into gradient formula ✓ answer (2)
3.3	$m_{LQ} \times 2 = -1 \quad [LQ \perp NS]$ $\therefore m_{LQ} = -\frac{1}{2}$ $-8 = -\frac{1}{2}(4) + c \quad \text{OR} \quad y + 8 = -\frac{1}{2}(x - 4)$ $c = -6 \quad \quad \quad y + 8 = -\frac{1}{2}x + 2$ $\therefore y = -\frac{1}{2}x - 6$	✓ m_{LQ} ✓ substitution of Q ✓ calculation of c or simplification (3)
3.4	OS is the radius of circle passing through S $(x - 0)^2 + (y - 0)^2 = (16)^2$ $x^2 + y^2 = 256$ Answer only: Full marks	✓ identifying radius = 16 ✓ Equation of circle (2)

3.5	$m_{RM} = m_{LQ} = -\frac{1}{2} \quad [RM \parallel LQ]$ $-4 = -\frac{1}{2}(6) + c \quad \text{OR} \quad y + 4 = -\frac{1}{2}(x - 6)$ $c = -1 \quad y + 4 = -\frac{1}{2}x + 3$ $\therefore y = -\frac{1}{2}x - 1$ $T(0; -1)$	$\checkmark m_{RM}$ $\checkmark$ substitution of $M(6; -4)$ $\checkmark$ coordinates of T (3)
3.6	$T(0; -1), P(0; -6) \text{ and } S(0; -16)$ $\therefore PS = 10 \text{ units and } TS = 15 \text{ units}$ $\frac{LS}{RS} = \frac{PS}{TS} = \frac{2}{3} \quad [\text{prop theorem; } RM \parallel LP]$ $\text{OR } [\text{line } \parallel \text{ one side of } \Delta / \text{lyn } \parallel \text{ een sy v } \Delta]$ Answer only: Full marks OR $M(6; -4), Q(4; -8) \text{ and } S(0; -16)$ $MS = \sqrt{180} = 6\sqrt{5} \text{ and } QS = \sqrt{80} = 4\sqrt{5}$ $\frac{LS}{RS} = \frac{QS}{MS} = \frac{2}{3} \quad [\text{prop theorem; } RM \parallel LQ]$ $\text{OR } [\text{line } \parallel \text{ one side of } \Delta / \text{lyn } \parallel \text{ een sy v } \Delta]$ Answer only: Full marks	$\checkmark PS = 10 \text{ units}$ $\checkmark TS = 15 \text{ units}$ $\checkmark$ answer (3) $\checkmark MS = 6\sqrt{5} \text{ units}$ $\checkmark QS = 4\sqrt{5} \text{ units}$ $\checkmark$ answer (3)
3.7	$\begin{aligned} \text{area of PTMQ} &= \text{area of } \Delta TSM - \text{area of } \Delta PSQ \\ &= \frac{1}{2} \cdot ST \cdot \perp h_M - \frac{1}{2} \cdot PS \cdot \perp h_Q \\ &= \frac{1}{2}(15)(6) - \frac{1}{2}(10)(4) \\ &= 45 - 20 \\ &= 25 \text{ square units} \end{aligned}$ OR $TM = \sqrt{45} = 3\sqrt{5} = 6,71$ $MQ = \sqrt{20} = 2\sqrt{5} = 4,47$ $PQ = \sqrt{20} = 2\sqrt{5} = 4,47$ $\begin{aligned} \text{area of trapezium PTMQ} &= \frac{1}{2}(3\sqrt{5} + 2\sqrt{5})(2\sqrt{5}) \\ &= \frac{1}{2}(5\sqrt{5})(2\sqrt{5}) \\ &= 25 \text{ square units} \end{aligned}$	$\checkmark$ area of $\Delta TSM -$ area of ΔPSQ $\checkmark$ area $\Delta TSM = 45$ $\checkmark$ area $\Delta PSQ = 20$ $\checkmark$ answer (4) $\checkmark TM = 3\sqrt{5}$ $MQ = 2\sqrt{5}$ $PQ = 2\sqrt{5}$ $\checkmark$ area of trapezium $= \frac{1}{2}$ (sum of sides)(height) $\checkmark$ substitute into formula $\checkmark$ answer (4)

	OR $MQ = \sqrt{20} = 2\sqrt{5}$ $PQ = \sqrt{20} = 2\sqrt{5}$ $TP = 5$ area of PTMQ = area of $\triangle MTP$ + area of $\triangle PQM$ $\text{area of PTMQ} = \frac{1}{2} TP \times \perp h_M + \frac{1}{2} MQ \times PQ$ area of PTMQ = $\frac{1}{2}(5) \times 6 + \frac{1}{2}(2\sqrt{5})(2\sqrt{5})$ area of PTMQ = $10 + 15 = 25$	✓ area of $\triangle MTP$ + area of $\triangle PQM$ ✓ area $\triangle MTP = 10$ ✓ area $\triangle PQM = 15$ ✓ answer (4)
		[19]

QUESTION 4



4.1	$PV = r = \sqrt{10}$ $PV = \sqrt{(k - (-3))^2 + (1 - 4)^2} = \sqrt{10}$ $(PV)^2 = (k - (-3))^2 + (1 - 4)^2 = 10$ $k^2 + 6k + 9 + 9 = 10$ OR $(k + 3)^2 + 9 = 10$ $k^2 + 6k + 8 = 0$ $(k + 3)^2 = 1$ $(k + 4)(k + 2) = 0$ $k + 3 = 1$ or $k + 3 = -1$ $k = -4$ or $k = -2$ $\therefore k = -2$	✓ $PV = r = \sqrt{10}$ ✓ substitution into distance formula ✓ standard form ✓ factors ✓ answer (5)
4.2	$x^2 + 6x + y^2 - 8y + 15 = 0$ y-intercepts: $(0)^2 + 6(0) + y^2 - 8y + 15 = 0$ $(y - 3)(y - 5) = 0$ $y_C = 3$ or $y_B = 5$ $\therefore BC = 2$ units	✓ $x = 0$ ✓ factors ✓ both values ✓ answer (4)

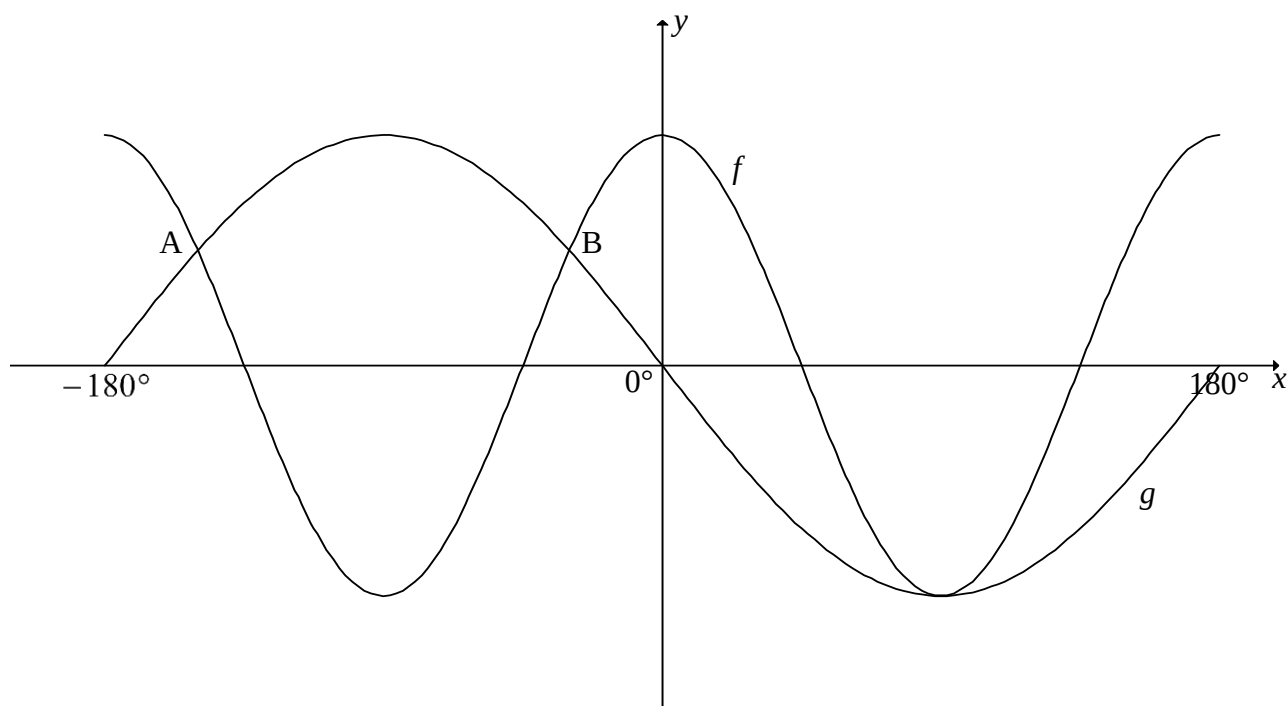
4.3.1	$m_{TC} = \frac{3-1}{0-(-2)}$ $= 1$ $\tan \alpha = 1$ $\therefore \alpha = 45^\circ$ OR $y = mx + 3$ $1 = m(-2) + 3$ $m_{TC} = 1$ $\tan \alpha = 1$ $\therefore \alpha = 45^\circ$	✓ substitution into gradient formula ✓ $\tan \alpha = 1$ ✓ answer (3)
4.3.2	$\hat{BCV} = 135^\circ$ [ext $\angle$ of Δ /buite $\angle$ v Δ] $\therefore \hat{VWB} = 45^\circ$ [opp $\angle$ s of cyclic quad/teenoorst. $\angle$ e v kvh] Answer only: Full marks OR $\hat{TCO} = 45^\circ$ [$\angle$ s of Δ / $\angle$ e v Δ] $\therefore \hat{VWB} = 45^\circ$ [ext $\angle$ s of cyclic quad/buite $\angle$ v kvh] Answer only: Full marks	✓ $\hat{BCV} = 135^\circ$ ✓ answer (2)
4.4.1	$Q(-3; -2)$	✓ x_Q ✓ y_Q (2)
4.4.2	$(x+3)^2 + (y+2)^2 = 10$	✓ LHS ✓ RHS (2)
4.4.3	$x = -2$ or $x = -4$	✓ $x = -2$ ✓ $x = -4$ (2)
		[20]

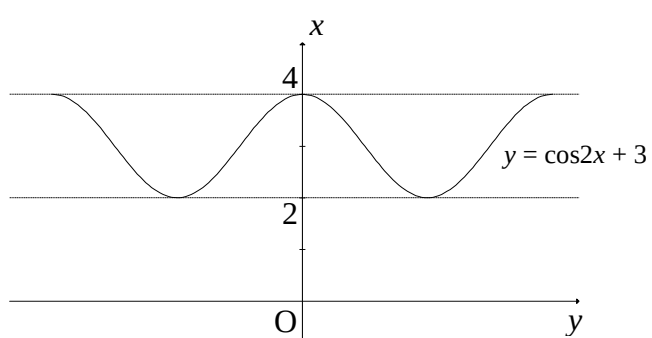
QUESTION/VRAAG 5

5.1	$\tan(-x) \cdot \cos x \cdot \sin(x - 180^\circ) - 1$ $= -\tan x \cdot \cos x \cdot \sin(-(180^\circ - x)) - 1$ $= \frac{-\sin x}{\cos x} \cdot \cos x \cdot (-\sin x) - 1$ $= \sin^2 x - 1$ $= -\cos^2 x$	$\checkmark -\tan x$ $\checkmark -\sin x \quad \checkmark \frac{-\sin x}{\cos x}$ $\checkmark \sin^2 x - 1$ $\checkmark \text{ answer}$ (5)
5.2.1	$\cos 215^\circ$ $= -\cos 35^\circ$ $= -m$	$\checkmark \text{ reduction}$ $\checkmark \text{ answer}$ (2)
5.2.2	$\sin 20^\circ$ $= \cos 70^\circ$ $= \cos 2(35^\circ)$ $= 2\cos^2 35^\circ - 1$ $= 2m^2 - 1$ OR $= \sin(55^\circ - 35^\circ)$ $= \sin 55^\circ \cos 35^\circ - \cos 55^\circ \sin 35^\circ$ $= m \cdot m - \sqrt{1-m^2} \cdot \sqrt{1-m^2}$ $= m^2 - (1-m^2)$ $= 2m^2 - 1$	$\checkmark \text{ co-function}$ $\checkmark \text{ double angle expansion}$ $\checkmark \text{ answer in terms of } m$ (3) $\checkmark \text{ compound angle expansion}$ $\checkmark \cos 55^\circ = \sqrt{1-m^2} \text{ or } \sin 35^\circ = \sqrt{1-m^2}$ $\checkmark \text{ answer in terms of } m$ (3)
5.3	$\cos 4x \cdot \cos x + \sin 4x \cdot \sin x = -0,7$ $\cos(4x - x) = -0,7$ $\text{ref } \angle = 45,57 \dots^\circ$ $3x = 180^\circ - 45,57 \dots^\circ + k \cdot 360^\circ \text{ or } 3x = 180^\circ + 45,57 \dots^\circ + k \cdot 360^\circ$ $3x = 134,43^\circ + k \cdot 360^\circ \text{ or } 3x = 225,57^\circ + k \cdot 360^\circ$ $x = 44,81^\circ + k \cdot 120^\circ; k \in \mathbb{Z} \quad x = 75,19^\circ + k \cdot 120^\circ; k \in \mathbb{Z}$	$\checkmark \text{ compound angle}$ $\checkmark 3x = 134,43^\circ \text{ or } 225,57^\circ$ $\checkmark x = 44,81^\circ \text{ or } 75,19^\circ$ $\checkmark + k \cdot 120^\circ; k \in \mathbb{Z}$ (4)

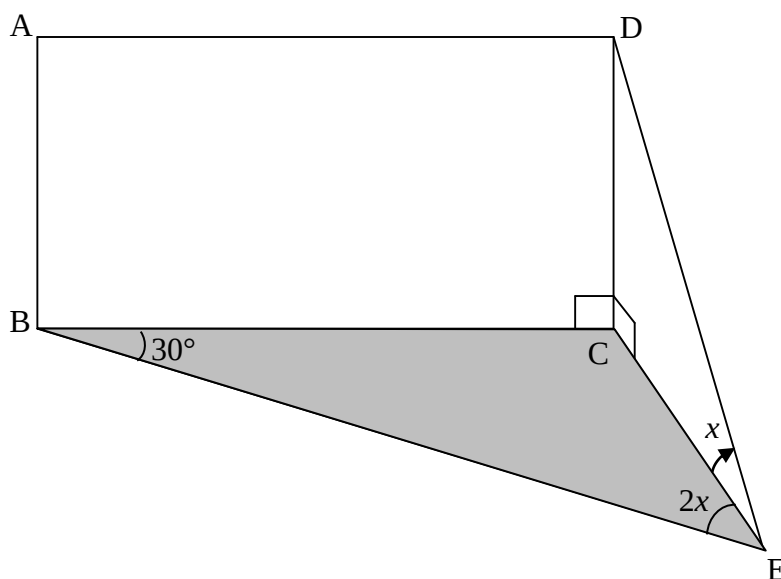
5.4	$\text{RHS} = \cos^2 x - \sin^2 x$ $\text{LHS} = \frac{\sin 4x \cdot \cos 2x - 2 \cos 4x \cdot \sin x \cdot \cos x}{\tan 2x}$ $= \frac{\sin 4x \cdot \cos 2x - \cos 4x \cdot \sin 2x}{\frac{\sin 2x}{\cos 2x}}$ $= \sin(4x - 2x) \left(\frac{\cos 2x}{\sin 2x} \right)$ $= \cos 2x$ $= \cos^2 x - \sin^2 x$ $\text{LHS} = \text{RHS}$	$\checkmark \sin 2x$ $\checkmark \frac{\sin 2x}{\cos 2x}$ $\checkmark \sin(4x - 2x)$ $\checkmark \cos 2x$
		(4)
		[18]

6.2



6.2.1	$A(-150^\circ; 0,5)$ $B(-30^\circ; 0,5)$ $AB = -30^\circ - (-150^\circ)$ $AB = 120^\circ$ Answer only: Full marks	$\checkmark AB = -30^\circ - (-150^\circ)$ $\checkmark$ answer (2)
6.2.2	$x \in (0^\circ; 90^\circ)$ or $x \in (90^\circ; 180^\circ)$ OR $0^\circ < x < 90^\circ$ or $90^\circ < x < 180^\circ$	$\checkmark x \in (0^\circ; 90^\circ)$ $\checkmark x \in (90^\circ; 180^\circ)$ (2) $\checkmark 0^\circ < x < 90^\circ$ $\checkmark 90^\circ < x < 180^\circ$ (2)
6.2.3	$\cos 2x = k - 3$ $k - 3 < -1$ or $k - 3 > 1$ $k < 2$ or $k > 4$ Answer only: Full marks OR  $k < 2$ or $k > 4$ Answer only: Full marks	$\checkmark k - 3 < -1$ or $k - 3 > 1$ $\checkmark k < 2$ $\checkmark k > 4$ (3) $\checkmark$ graph of $y = \cos 2x + 3$ $\checkmark k < 2$ $\checkmark k > 4$ (3)
[13]		

QUESTION/VRAAG 7



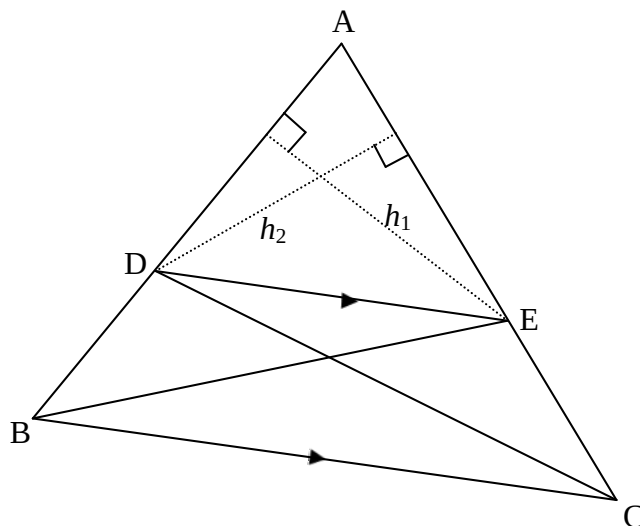
7.1	In $\triangle BCE$: $\frac{CE}{\sin \hat{B}} = \frac{BC}{\sin \hat{E}}$ $\frac{CE}{\sin 30^\circ} = \frac{BC}{\sin 2x}$ $CE = \frac{BC \sin 30^\circ}{\sin 2x}$ In $\triangle CDE$: $\frac{DC}{CE} = \tan \hat{D}$ $DC = \frac{BC \cdot \sin 30^\circ}{\sin 2x} (\tan x)$ $DC = \frac{BC}{4 \sin x \cos x} \left(\frac{\sin x}{\cos x} \right)$ $DC = \frac{BC}{4 \cos^2 x}$	✓ correct use of sine rule ✓ $CE = \frac{BC \sin 30^\circ}{\sin 2x}$ ✓ correct trig ratio ✓ Subst CE ✓ $2 \sin x \cos x$ ✓ $\frac{\sin x}{\cos x}$ (6)
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7.2	$DC = \frac{BC}{4 \cos^2 30^\circ}$ $= \frac{BC}{4 \left(\frac{\sqrt{3}}{2} \right)^2}$ $= \frac{BC}{3}$ $\therefore BC = 3DC$ But $AB = DC$ [opp sides of rectangle/teenoorst. sye v reghoek] $\therefore BC = 3AB$ Area of rectangle = $(AB)(BC)$ $= (AB)(3AB)$ $= 3AB^2$	$\checkmark DC = \frac{BC}{3}$ $\checkmark BC = 3AB$ $\checkmark \text{substitution into area formula}$ (3)
[9]		

	OR $\hat{C}_3 = 3x$ [ext $\angle$ of cyclic quad/<i>buite</i> $\angle$ v <i>kvh</i>] $\hat{D}_1 = 4x$ [ext $\angle$ of Δ/<i>buite</i> $\angle$ v Δ] $2x + 3x + 4x = 180^\circ$ [sum of $\angle$s in Δ/<i>∠e</i> v Δ] $9x = 180^\circ$ $x = 20^\circ$	✓ S ✓R ✓ S ✓ S ✓R ✓ answer (6)
[16]		

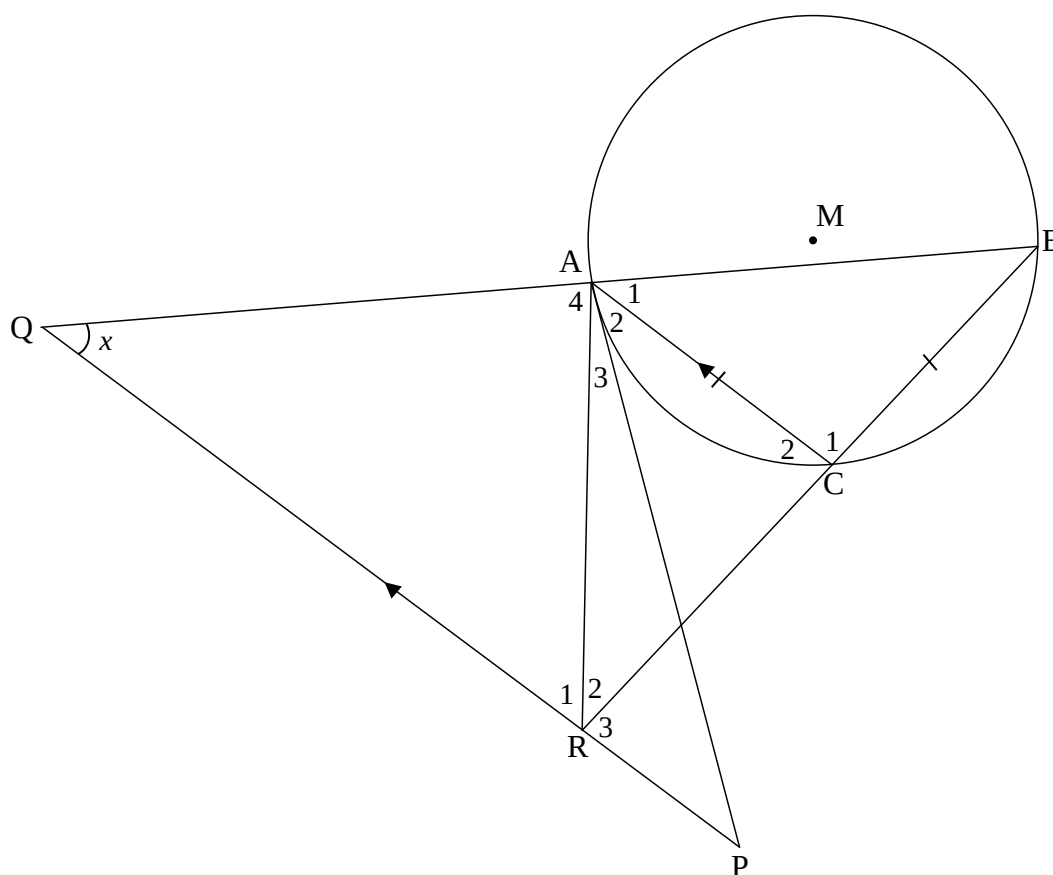
QUESTION/VRAAG 9

9.1



9.1	Constr: Join BE and CD and draw h_1 from E $\perp$ AD and h_2 from D $\perp$ AE Konstr: Verbind BE en CD en trek h_1 vanaf E $\perp$ AD en h_2 vanaf D $\perp$ AE Proof/Bewys: $\frac{\text{area } \triangle ADE}{\text{area } \triangle BDE} = \frac{\frac{1}{2}AD \times h_1}{\frac{1}{2}BD \times h_1} = \frac{AD}{BD}$ $\frac{\text{area } \triangle ADE}{\text{area } \triangle DEC} = \frac{\frac{1}{2}AE \times h_2}{\frac{1}{2}EC \times h_2} = \frac{AE}{EC}$ area $\triangle ADE$ = area $\triangle ADE$ [common/<i>gemeenskaplik</i>] But area $\triangle BDE$ = area $\triangle DEC$ [same base & height ; DE BC/ <i>dies basis & hoogte ; DE BC</i>] $\therefore \frac{\text{area } \triangle ADE}{\text{area } \triangle BDE} = \frac{\text{area } \triangle ADE}{\text{area } \triangle DEC}$ $\therefore \frac{AD}{BD} = \frac{AE}{EC}$	✓ constr/konstr $\checkmark \frac{\text{area } \triangle ADE}{\text{area } \triangle BDE} = \frac{\frac{1}{2}AD \times h_1}{\frac{1}{2}BD \times h_1} \quad \textbf{or R}$ $\checkmark \frac{\text{area } \triangle ADE}{\text{area } \triangle DEC} = \frac{AE}{EC}$ ✓ S ✓R
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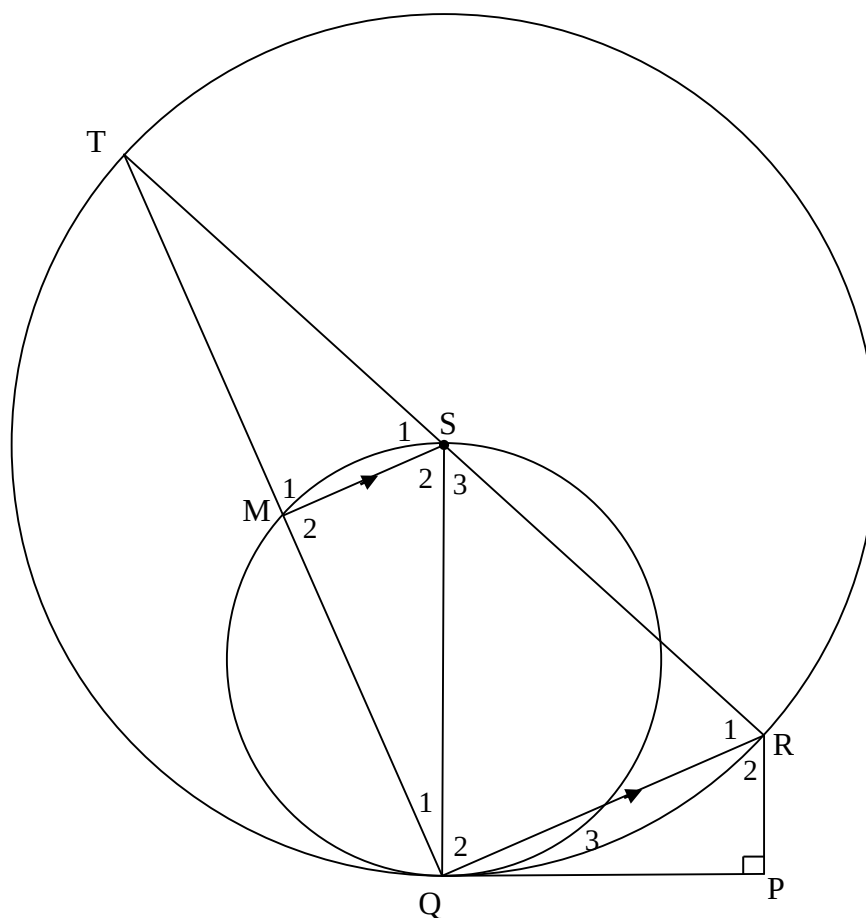
9.2



9.2.1	$\hat{A}_1 = x$ [corresp $\angle$ s; $PQ \parallel CA$ /ooreenkomstige $\angle$ e, $PQ \parallel CA$] $\hat{B} = x$ [$\angle$ s opp equal sides/ $\angle$ e teenoor gelyke sye] $\hat{A}_2 = x$ [tan-chord theorem/ $\angle$ tussen raaklyn en koord] $\hat{P} = x$ [alt $\angle$ s; $PQ \parallel CA$ /verw. $\angle$ e, $PQ \parallel CA$]	$\checkmark$ S $\checkmark$ R $\checkmark$ S/R $\checkmark$ S $\checkmark$ R $\checkmark$ S/R	(6)
9.2.2	$\hat{B} = \hat{P}$ [proved in 9.2.1/bewys in 9.2.1] $\therefore$ A, B, P and R are concyclic $\therefore$ ABPR is a cyclic quadrilateral [conv $\angle$ s in the same segment/ koord onderspan gelyke omtreks $\angle$ e]	$\checkmark$ S $\checkmark$ R	(2)
9.2.3	$\frac{BA}{BQ} = \frac{BC}{BR}$ [prop th; $AC \parallel QP$] OR [line $\parallel$ one side Δ /lyn $\parallel$ een syn v Δ] But $QR = BR$ [sides opp = $\angle$ s/sye teenoor = $\angle$ e] $\therefore \frac{BA}{BQ} = \frac{BC}{QR}$	$\checkmark$ S $\checkmark$ R $\checkmark$ S	(3)

	OR In $\triangle ABC$ and $\triangle BQR$: $\hat{A}_1 = \hat{B} = x \quad [\text{proved in 9.2.1}]$ $\hat{B} = \hat{Q} = x \quad [\text{proved in 9.2.1}]$ $\hat{C}_1 = \hat{B}RQ = 180^\circ - 2x \quad [\text{sum of } \angle\text{s of } \triangle]$ $\therefore \triangle ABC \parallel \triangle BQR$ $\therefore \frac{BA}{BQ} = \frac{BC}{QR}$ OR In $\triangle ABC$ and $\triangle BQR$: $\hat{A}_1 = \hat{B} = x \quad [\text{proved in 9.2.1}]$ $\hat{B} = \hat{Q} = x \quad [\text{proved in 9.2.1}]$ $\hat{C}_1 = \hat{B}RQ = 180^\circ - 2x \quad [\text{sum of } \angle\text{s of } \triangle]$ $\therefore \triangle ABC \parallel \triangle BQR \quad [\angle\angle\angle]$ $\therefore \frac{BA}{BQ} = \frac{BC}{QR}$ OR In $\triangle ABC$ and $\triangle QBR$: $\hat{B}$ is common $\hat{A}_1 = \hat{Q} = x \quad [\text{corres } \angle\text{s; } PQ \parallel CA]$ $\hat{C}_1 = \hat{B}RQ = 180^\circ - 2x \quad [\text{sum of } \angle\text{s of } \triangle]$ $\therefore \triangle ABC \parallel \triangle QBR \quad [\angle\angle\angle]$ But $QR = BR$ [sides opp = $\angle$s/sye teenoor = $\angle$e] $\therefore \frac{BA}{BQ} = \frac{BC}{QR}$	✓ S ✓ S ✓ S (3) ✓ S ✓ S ✓ R (3) ✓ S ✓ S ✓ S (3)
[17]		

QUESTION/VRAAG 10



10.1.1	$\hat{Q}_1 + \hat{Q}_2 = 90^\circ$	[$\angle$ in semi circle/ $\angle$ in halwe sirkel]	✓ S/R	
	$\therefore \hat{M}_2 = 90^\circ$	[co-interior $\angle$, $MS \parallel QR$ /ko-binne $\angle$ e, $MS \parallel QR$]	✓ S/R	
	$\therefore SQ$ is a diameter	[converse: $\angle$ in semi circle/ Omgekeerde: $\angle$ in halwe sirkel]	✓ R	
	OR			(3)
	$MS \parallel QR$			
	$\frac{TS}{SR} = \frac{TM}{MQ} = \frac{1}{1}$	[prop theorem; $SM \parallel QR$] OR [line $\parallel$ one side of Δ]/lyn $\parallel$ een sy v Δ	✓ S/R	
	$\therefore TM = MQ$			(3)
	$\therefore \hat{M}_2 = 90^\circ$	[Line from centre bisects chord/midpt. sirkel; midpt koord]	✓ S/R	
	$\therefore SQ$ is a diameter	[converse: $\angle$ in semi circle/ Omgekeerde: $\angle$ in halwe sirkel]	✓ R	
	OR			(3)
	$SQ \perp QP$	[tan $\perp$ rad/raaklyn $\perp$ radius]	✓ S ✓ R	
	$\therefore SQ$ is a diameter	[converse: tan $\perp$ rad/Omgekeerde: raaklyn $\perp$ radius]	✓ R	

10.1.2	In $\triangle RTQ$ and $\triangle RQP$ $\hat{T} = \hat{Q}_3$ [tan-chord theorem/$\angle$ tussen raaklyn en koord] $\hat{Q}_1 + \hat{Q}_2 = 90^\circ$ [co-interior $\angle$s, $MS \parallel QR$/ko-binne $\angle$e, $MS \parallel QR$] or [$\angle$ in semi circle/$\angle$ in halwe sirkel] $\therefore \hat{Q}_1 + \hat{Q}_2 = \hat{P} = 90^\circ$ $\hat{R}_1 = \hat{R}_2$ [$\angle$s of $\triangle$/$\angle$e van $\triangle$] $\triangle RTQ \parallel \triangle RQP$ $\frac{RT}{RQ} = \frac{RQ}{RP}$ $RT = \frac{RQ^2}{RP}$ OR In $\triangle RTQ$ and $\triangle RQP$ $\hat{T} = \hat{Q}_3$ [tan-chord theorem $\angle$ tussen raaklyn en koord] $\hat{Q}_1 + \hat{Q}_2 = 90^\circ$ [co-interior $\angle$s, $MS \parallel QR$/ko-binne $\angle$e, $MS \parallel QR$] or [$\angle$ in semi circle/$\angle$ in halwe sirkel] $\therefore \hat{Q}_1 + \hat{Q}_2 = \hat{P} = 90^\circ$ $\triangle RTQ \parallel \triangle RQP$ [$\angle, \angle, \angle$] $\frac{RT}{RQ} = \frac{RQ}{RP}$ $RT = \frac{RQ^2}{RP}$	✓ S ✓ R ✓ S ✓ S ✓ S ✓ ratio (6) ✓ S ✓ R ✓ S ✓ S ✓ R ✓ ratio (6)
10.2	$QR = 28$ units [midpoint theorem/midpt. stelling] $RP^2 = 28^2 - (\sqrt{640})^2$ [Pythagoras/Pythagoras] $RP = 12$ units $RT = \frac{RQ^2}{RP}$ $RT = \frac{28^2}{12}$ $RT = \frac{196}{3}$ Radius = $\frac{98}{3}$ units	✓ S ✓ R ✓ S ✓ $RP = 12$ ✓ RT ✓ answer (6)
		[15]

TOTAL/TOTAAL: 150